

Internet Search Engines

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Abstract—Internet search engines have become such an integral part in the use of the Internet as a whole that search engines can be determined to be just as important as other aspects of the world wide web. Internet search engines can be broken down to three key components, which are the web crawlers, indexing program, and the retrieval software. In this paper, I will be discussing the improvement in image searching through web crawler enhancements.

1 PROBLEM SPACE

1.1 Problem and Context

Currently, Internet search engines typically utilize keywords to search throughout the world wide web to return relevant results. However, image searching, which utilizes an image instead of keywords as the context to search has become very frequently used. An issue that many others and myself have come across is the lack of accuracy in the returned results. There are many aspects that have led to this problem becoming widespread, which includes the lack of accuracy in the image reading software and the web crawling logic. Google's reverse image search is a great example to look at.

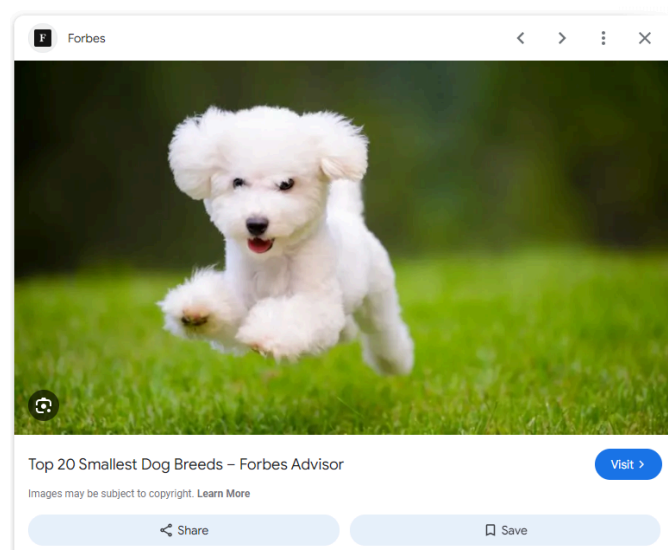


Figure 1—Google Search Image of Dog. [Source](#)

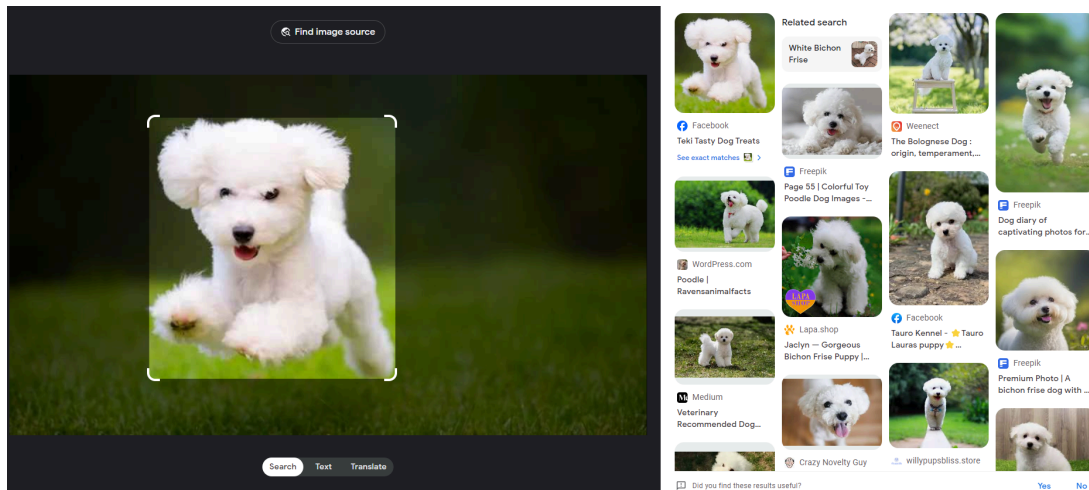


Figure 1—Google’s Reverse Search Image of above image. [Source](#)

The image above is of a random dog that I have searched on Google, and the image itself is taken from Forbes. The second image below the first is the results from Google’s reverse image search. As you can see, the search engine’s results show a multitude of similar dogs, and there is even one search result that is exactly the same image. However, if you look closer, that search result is not from the original Forbes site, but instead from Facebook. The original site is not shown within the search results. This simply demonstrates that although the search engine was able to give similar results that can be somewhat acceptable, it is not up to the level that should be expected for a search engine image search, especially if it is not able to even return the original post that the image was from. The improvement that I am suggesting for search engines that utilize images is the redesigning of the current web crawling logic behind the image search.

1.2 Proposed Design

The web crawling of the image search currently works almost exactly in the same way a search engine would give relevant results for a text search (How Does Picture Search Work? | Algolia, n.d). The only difference is that the image itself is used in conjunction with generated keywords from the image for the search. For example, with the image of the dog before, some possible key words would be dog, white, grass, and running. The actual analysis of the image itself, which involves breaking down the image and feeding it into machine learning tools to generate texts of what the image is. These texts would then be used in the typical search engine for web crawling to get the usual results. My design revolves around including an image comparison program within the web crawler to be able to give accurate comparison of images found using the text search results. The process of the web crawler will not change at all except for the addition of this new process. Whenever the web crawler crawls and comes across an image, both images will be compared for possible similarities utilizing a separate machine learning tool to determine

if the images are similar. As long as the image has at least a 30% similarity in the image itself then this url will be added to the return result. This will ultimately ensure that despite changes to image cropping, shading, or other possible manipulation, the search engine will always return results that have the original image within the results.

1.3 Comparison

This redesign of the current reverse image search engines ultimately allows users to search for specific images instead of just getting possible similar images on the Internet. One of the main reasons people use reverse image search is to be able to find exact copies of his or her image online (Golowczynski, 2020). If someone wanted to find similar pictures he or she could simply just give text descriptions instead. Image searches are rarely used; however, if one of its main features is not as accurate as it can be then reverse image search engines will never be able to grow.

2 NEED FINDING EXERCISE

2.1 Need Finding

Similarly to the example above, this example utilizes a basic photo of a sunflower found on Google. Then the image is color shifted for the search. However, when put through the reverse image search, the results did not show the original post at all.

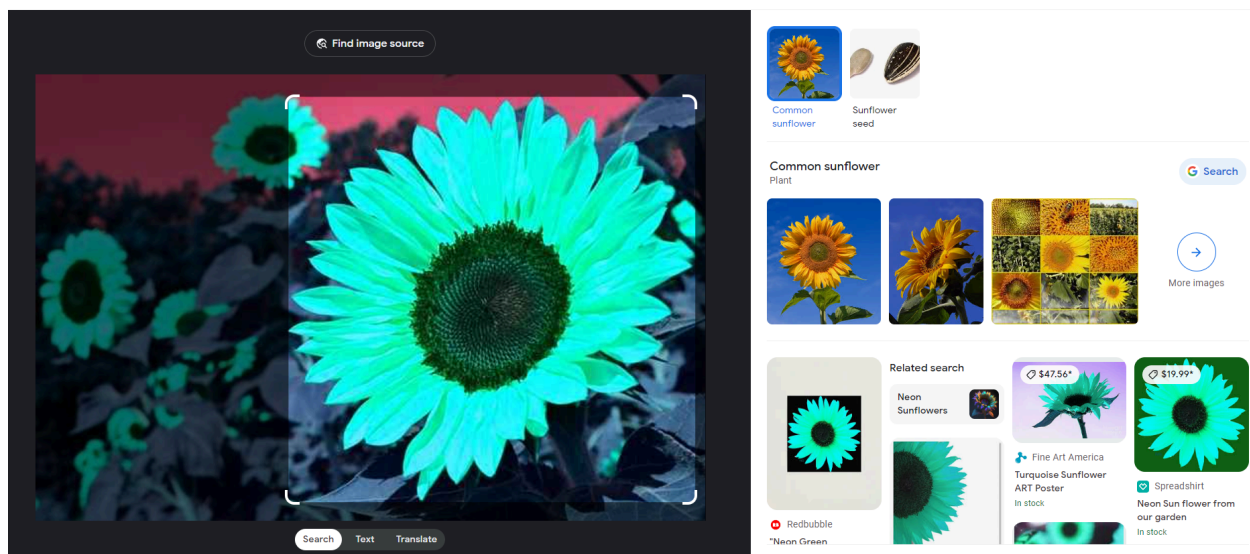


Figure 3—Google Search Image of Sunflower. [Source](#)

2.2 Functional Components

For this specific design plan, the main change to the original reverse image search engine is the addition of the machine learning image comparison for images found when web crawling. This machine learning tool will be trained using the original image to be able to identify the image whether it has been cropped or not. Then it is simply used in conjunction with the web crawler so that every image the web crawler comes across is tested with the machine to determine if the image is the same or not.

2.3 Architecture

Pseudo Code of Machine Learning Tool for Cropping to be added to web crawler

ML Training Process

Initialize Machine and input original image

For $i = 0$ til $i = \text{length of image}$

 Input cropped image of length of image - i

 Input cropped image of width of image - i

 Input cropped image of length of image - i and width of image - i

For $i = \text{length of image}$ til $i = 0, -1$

 Input cropped i - image of length of image

 Input cropped i - image of width of image

 Input cropped i - image of length of image and i - image of width of image

Once training is done add to web crawler

While web crawler encounters image

 Test image with machine

 If returned data of at least 0.3 add to indexing software

2.4 Data Collection

For this design, the main data collection is completely geared towards the accuracy of the returned search results. This means that the data that needs to be recorded is mainly the search results returned to the user so that it can be evaluated. In addition, the image and text query itself

will also need to be logged in the database. Finally, the time it takes for each query to crawl needs to also be tracked for any efficiency issues.

2.5 Utility Types

With the addition of this functionality, users will be able to have an easier time trying to find similar or exact copies of images through the reverse image search engine. The main goal of this change to the web crawler is to be able to better give results that have exact images to its users in addition to returning results of similar images.

3 DESIGN REFINEMENTS

3.1 Need Finding

For this need finding the issues also include cropped photos not just color shifted photos.

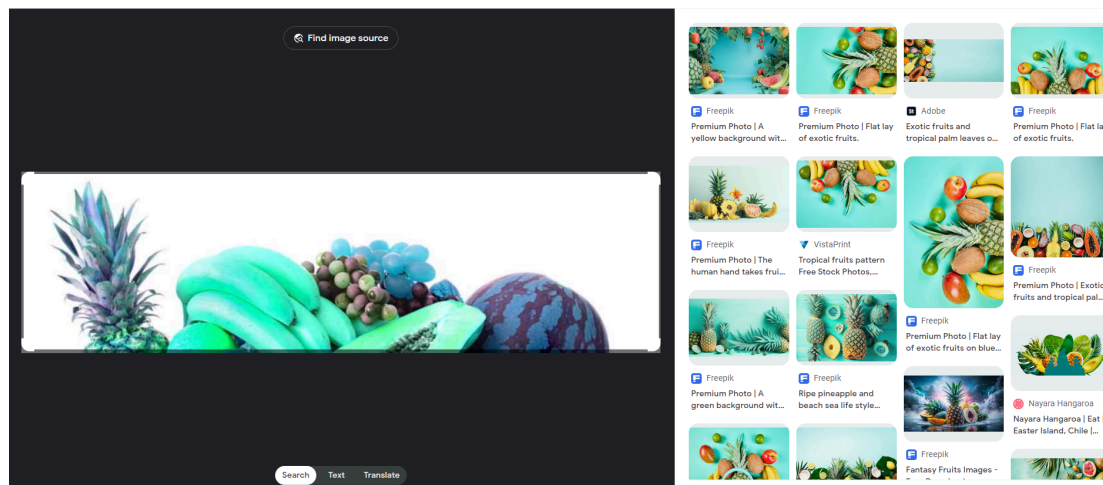


Figure 4—Google Reverse Image Search of Color Shifted and Cropped Fruit. [Source](#)

3.2 Functional Components

Similarly to the previous design plan, this design also includes the color shifting aspect of images. In addition, it will also handle issues such as cropped images in the search engine. The machine learning tool that will be included with the web crawler will handle both of these issues when testing the images found during web crawling to also be able to identify manipulated photos that were once the original.

3.3 Architecture

Pseudo Code of Machine Learning Tool for Color Shift and Cropping to be added to web crawler

ML Training Process

Initialize Machine and input original image

For $i = 0$ til $i = 510$

input color shifted image by $(x+1,y,z)$, $(x,y+1,z)$, $(x,y,z+1)$ of every pixel in the image for x,y,z between -255 and 255

For $i = 0$ til $i = \text{length of image}$

Input cropped image of length of image - i

Input cropped image of width of image - i

Input cropped image of length of image - i and width of image - i

For $i = \text{length of image}$ til $i = 0, -1$

Input cropped i - image of length of image

Input cropped i - image of width of image

Input cropped i - image of length of image and i - image of width of image

Once training is done add to web crawler

While web crawler encounters image

Test image with machine

If returned data of at least 0.3 add to indexing software

3.4 Data Collection

Similarly to the previous design, this refined design requires the data collection of the returned user's results so that the data can be analyzed. This strongly supports a database that can keep track of the image, the text query, and at least the first 100 returned results of the search engine. In addition, a major point to consider is the speed of the web crawling since the image comparison can slow down the crawling, so the database will need to be able to keep track of each query as a whole.

3.5 Utility Types

The intended purpose of this design focuses on fixing the inaccuracies that can come from image manipulation that can affect the results of image search engines. These designs are geared towards all types of users so that it is done completely in the background whenever anyone is using reverse image searching. The web crawling will be done completely as normal just with

the addition of the image comparison, which will ultimately improve the results that are given back to the user.

4 ADDITIONAL DESIGN CONSIDERATIONS

4.1 Evaluation of Baseline

The evaluation metrics of the designs are mainly subjective when it comes to accuracy of the image search results and empirical when it comes to efficiency in the speed of the web crawling. The time of each web crawl for the queries will be evaluated to see if the increased time from the additional image comparison will severely affect the speed of the search engine or not. In addition, the accuracy of the image search will need to be evaluated to see if the images are able to properly pinpoint exact copies or altered images from the results. As long as there is not a significant increase in time and there are positive image search results supporting better accuracy in the search engine then the design can be considered to be a better system performance overall.

4.2 Additional Functional Requirements

For the baseline design approach, it is highly specified towards the accuracy and functionality of the image search engine as a whole. However, this will lead to possible lack of other important aspects overall. Further functional requirements that will definitely be evaluated and improved would be the efficiency and flexibility of the image search engine. With the changes from the baseline design, efficiency is very crucial to at least keep it the same as before so that users do not have to wait a long time for their results. In addition, another possible addition would be allowing users to combine image and text search while querying in the search engine. This will give users even more flexibility in the use of the search engine.

5 REFERENCES

1. *How does picture search work?* | Algolia. (n.d.). Algolia Blog. <https://www.algolia.com/blog/product/picture-search-how-does-an-image-finder-search-engine-work/>
2. Golowczynski, M. (2020, November 13). *Google Reverse Image Search: Everything You Need to Know*. SmartFrame. <https://smartframe.io/blog/google-reverse-image-search-everything-you-need-to-know/>